MNRAS 000, 1–3 (2026) Preprint 21 July 2026 Compiled using MNRAS LATEX style file v3.0

PLACEHOLDER TITLE: Testing the Molten Gas Dwarf Hypothesis for K2-18 b with Coupled Interior-Atmosphere Evolution Models

Robb Calder,[1★] Oliver Shorttle,[1,2] Harrison Nicholls,[1,3] Tim Lichtenberg,[4] Claire Marie Guimond[3,5]
[1]*Institute of Astronomy, University of Cambridge, Madingley Road, Cambridge CB3 0HA, UK*
[2]*Department of Earth Sciences, University of Cambridge, Downing Street, Cambridge CB2 3EQ, UK*
[3]*Atmospheric, Oceanic, and Planetary Physics, Department of Physics, University of Oxford, Oxford OX1 3PU, United Kingdom*
[4]*Kapteyn Astronomical Institute, University of Groningen, P.O. Box 800, 9700 AV Groningen, The Netherlands*
[5]*Department of Earth and Planetary Sciences, ETH Zurich, Sonneggstrasse 5, 8092 Zurich, Switzerland*

Draft in preparation

ABSTRACT

PLACEHOLDER ABSTRACT. K2-18 b is a sub-Neptune whose observed transmission spectrum has been proposed as evidence for a Hycean world with a liquid-water ocean beneath an H_2-dominated atmosphere. An alternative interpretation is that K2-18 b is a 'gas dwarf' – a planet with a silicate/iron interior and an H_2-dominated envelope – with a permanent magma ocean in contact with its atmosphere. Volatile exchange between such a magma ocean and its overlying atmosphere would imprint a distinct chemical signature on the observed spectrum. In this work we use the coupled interior-atmosphere evolution model `PROTEUS` to generate self-consistent synthetic transmission spectra for K2-18 b across a grid spanning mantle oxygen fugacity and the bulk volatile hydrogen, carbon, nitrogen and sulphur budgets, and compare these synthetic spectra against the observed transmission spectrum/spectra of K2-18 b. We assess whether the observations of K2-18 b are consistent with the molten gas dwarf hypothesis. [SUMMARISE KEY RESULT HERE.] [STATE CONCLUSION HERE.]

Key words: planets and satellites: interiors – planets and satellites: physical evolution – planets and satellites: atmospheres – planets and satellites: composition – planets and satellites: individual: K2-18 b

1 INTRODUCTION

PLACEHOLDER. Introduce the sub-Neptune population and the observational degeneracy between interior structures consistent with a given mass and radius (Fulton et al.2017; Rogers et al.2023). Introduce K2-18 b specifically: its system parameters, its JWST transmission spectrum (Madhusudhan et al.2023, 2025; Hu et al.2025), and the 'Hycean world' interpretation of that spectrum (Madhusudhan et al.2021, 2023). Introduce the competing 'gas dwarf' interpretation, under which a magma ocean in contact with an H_2-dominated atmosphere could produce a similar observed spectrum via volatile exchange with the melt (Shorttle et al.2024; Rigby et al.2024; Wogan et al.2024). State the aim of this paper: to test the molten gas dwarf hypothesis for K2-18 b directly, by comparing observations against self-consistent synthetic spectra generated from an evolutionary interior-atmosphere model across a grid in mantle redox state and bulk volatile (H, C, N, S) budget, rather than by mass-radius-instellation arguments alone (cf. Cloutier et al.2017; Sarkis et al.2018; Benneke et al.2019). Close with a paragraph outlining the structure of the paper.

★ E-mail: rdc49@cam.ac.uk

2 METHODS

2.1 Model Description

PLACEHOLDER. Briefly describe the `PROTEUS` coupled interior-atmosphere evolution framework: the interior thermal evolution module (`SPIDER`/`ARAGOG`), the radiative-convective climate module (`AGNI`, backed by `SOCRATES`), and the volatile outgassing/partitioning module (`CALLIOPE`) (Lichtenberg et al.2021; Nicholls et al.2024, 2025b,a; Bower et al.2022). State that each simulation starts fully molten and evolves until either the mantle solidifies or the planet reaches global energetic steady state (the permanent magma ocean outcome), and that atmospheric chemistry and the synthetic transmission spectrum for each simulation are produced automatically as part of the PROTEUS pipeline. Reference `CLAUDE.md` in the project root for the technical details of how PROTEUS output for this project is organised and where synthetic spectra live within each run's output directory.

2.2 K2-18 b System Parameters

PLACEHOLDER. State the fiducial K2-18 b system parameters adopted for this study (planet mass, planet radius, orbital period/semi-major axis, instellation flux, host star spectral type/effective temperature/luminosity), with citations to the relevant discovery/characterisation papers (Cloutier et al.2017; Sarkis et al.2018; Benneke et al.2019), in Table 1.

Parameter	Value	Source
Planet Mass ($M_\oplus$)	TBD	TBD
Planet Radius ($R_\oplus$)	TBD	TBD
Orbital Period (d)	TBD	TBD
Instellation Flux ($F_\oplus$)	TBD	TBD
Host Star $T_{\rm eff}$ (K)	TBD	TBD
Host Star Spectral Type	TBD	TBD

Table 1. PLACEHOLDER. Fiducial K2-18 b system parameters adopted in this study.

2.3 Fiducial Model Parameters

PLACEHOLDER. List the PROTEUS model parameters held fixed across the grid sweep (e.g. core radius fraction, core density/heat capacity, boundary layer properties, rheological transition parameters, radiogenic heating, spectral/vertical resolution), following the same style of table as used in the predecessor population study, in Table 2.

2.4 Parameter Grid: Mantle Redox State and Volatile Budgets

PLACEHOLDER. Describe the grid swept in this study: mantle oxygen fugacity (relative to a redox buffer) and the bulk volatile hydrogen, carbon, nitrogen and sulphur budgets, including the sampled range/values for each axis and how the grid points were generated with `proteus grid`. Justify the chosen ranges relative to plausible bulk compositions for gas dwarfs (Wolfgang & Lopez2015; Ginzburg et al.2016; Tang et al.2024). Note which grid points have completed and been checked into `simulation_data/` at the time of writing.

2.5 Synthetic Spectra and Observational Comparison

PLACEHOLDER. Describe how the synthetic transmission spectrum is generated for each completed grid point as part of the PROTEUS pipeline, the observed K2-18 b transmission spectrum/spectra used for comparison (source and reduction pipeline), and the statistical method used to compare synthetic and observed spectra (e.g. χ^2, retrieval, or goodness-of-fit metric) to assess consistency with the molten gas dwarf hypothesis.

3 RESULTS

3.1 Grid Overview

PLACEHOLDER. Summarise the outcome of the grid sweep: which regions of the fO2-H-C-N-S parameter space yield a permanent magma ocean versus a solidified mantle for the K2-18 b system parameters, and the resulting range of atmospheric compositions.

3.2 Comparison with the Observed Spectrum

PLACEHOLDER. Present the comparison between synthetic spectra across the grid and the observed K2-18 b spectrum/spectra, identifying which regions of parameter space (if any) are statistically consistent with the observations.

3.3 Sensitivity to Mantle Redox State and Volatile Budgets

PLACEHOLDER. Discuss how the goodness-of-fit to the observed spectrum varies across the fO2, H, C, N and S axes of the grid, and which of these parameters most strongly control the synthetic spectrum.

4 DISCUSSION

PLACEHOLDER. Discuss the results in the context of the Hycean world interpretation of K2-18 b (Madhusudhan et al.2021, 2023; Hu et al.2025) and other gas-dwarf/magma-ocean interpretations (Shorttle et al.2024; Rigby et al.2024; Wogan et al.2024). Discuss caveats (e.g. model assumptions, stellar spectrum choice, atmospheric escape, retrieval degeneracies) and what additional observations or modelling would help discriminate between hypotheses for K2-18 b.

5 CONCLUSIONS

PLACEHOLDER. Summarise whether the observations of K2-18 b are, or are not, consistent with it being a molten gas dwarf, and state the key implications and recommended follow-up work.

ACKNOWLEDGEMENTS

PLACEHOLDER.

DATA AVAILABILITY

PLACEHOLDER. State how the PROTEUS grid-sweep output and plotting scripts underlying this paper's figures will be made available.

REFERENCES

Benneke B., et al., 2019, ApJ, 887, L14

Bower D. J., Hakim K., Sossi P. A., Sanan P., 2022, The Planetary Science Journal, 3, 93

Cloutier R., et al., 2017, A&A, 608, A35

Fulton B. J., et al., 2017, AJ, 154, 109

Ginzburg S., Schlichting H. E., Sari R., 2016, ApJ, 825, 29

Hu R., et al., 2025, arXiv e-prints, p. arXiv:2507.12622

Lichtenberg T., Bower D. J., Hammond M., Boukrouche R., Sanan P., Tsai S.-M., Pierrehumbert R. T., 2021, Journal of Geophysical Research (Planets), 126, e06711

Madhusudhan N., Piette A. A. A., Constantinou S., 2021, ApJ, 918, 1

Madhusudhan N., Sarkar S., Constantinou S., Holmberg M., Piette A. A. A., Moses J. I., 2023, ApJ, 956, L13

Madhusudhan N., Constantinou S., Holmberg M., Sarkar S., Piette A. A. A., Moses J. I., 2025, ApJ, 983, L40

Nicholls H., Lichtenberg T., Bower D. J., Pierrehumbert R., 2024, Journal of Geophysical Research (Planets), 129, 2024JE008576

Nicholls H., Pierrehumbert R., Lichtenberg T., 2025a, The Journal of Open Source Software, 10, 7726

Nicholls H., Pierrehumbert R. T., Lichtenberg T., Soucasse L., Smeets S., 2025b, MNRAS, 536, 2957

Rigby F. E., et al., 2024, ApJ, 975, 101

Rogers J. G., Schlichting H. E., Owen J. E., 2023, ApJ, 947, L19

Sarkis P., et al., 2018, AJ, 155, 257

Shorttle O., Jordan S., Nicholls H., Lichtenberg T., Bower D. J., 2024, ApJ, 962, L8

Parameter Name	Fiducial Value	Justification for Fiducial Value
Planet Mass	TBD	K2-18 b system value (Table 1)
Planet Instellation	TBD	K2-18 b system value (Table 1)
Stellar $T_{\rm eff}$	TBD	K2-18 b system value (Table 1)
Core Radius Fraction	TBD	TBD
Core Density	TBD	TBD
Core Heat Capacity	TBD	TBD
Boundary Layer Conductivity	TBD	TBD
Boundary Layer Thickness	TBD	TBD
Number of Spectral Bands	TBD	TBD
Number of Atmospheric Levels	TBD	TBD
Number of Interior Levels	TBD	TBD

Table 2. PLACEHOLDER. Fiducial `PROTEUS` model parameters held constant across the grid sweep.

Tang Y., Fortney J. J., Nimmo F., Thorngren D., Ohno K., Murray-Clay R., 2024, arXiv e-prints, p. arXiv:2410.21584
Wogan N. F., Batalha N. E., Zahnle K. J., Krissansen-Totton J., Tsai S.-M., Hu R., 2024, ApJ, 963, L7
Wolfgang A., Lopez E., 2015, ApJ, 806, 183

APPENDIX A: MODEL PARAMETERS AND SOLUBILITY LAWS

PLACEHOLDER. Detail the solubility laws and thermochemical equilibrium coefficients used by `CALLIOPE` for each volatile species tracked in this study, following the approach of the predecessor population study.

APPENDIX B: ADDITIONAL GRID DIAGNOSTICS

PLACEHOLDER. Any supplementary figures/tables diagnosing the grid sweep (e.g. atmospheric mass, mean molecular weight, or speciation as a function of grid parameters) that support but are not central to the main results.

This paper has been typeset from a TEX/LATEX file prepared by the author.